



JNANA KARRI

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PROFESSIONAL SUMMARY

Data Engineer with 7+ years owning and scaling cloud-native data platforms on AWS. Expertise in ETL/ELT pipelines, data modeling, and distributed systems, delivering high-quality datasets, reporting solutions, and analytics systems for business decision-making.

AWS Certified Solutions Architect – Professional (4× AWS Certified) with strong experience in system design, cloud architecture, focused on building trusted, high-quality datasets with strong data governance and production-grade data platforms.

SKILLS

- Languages:** Python, SQL
- Data:** ETL/ELT, Data Modeling, Data Warehousing, Lakehouse
- Big Data:** Spark, PySpark
- AWS:** Glue, Redshift, S3, Athena, Lambda, Step Functions, Kinesis
- Orchestration/(CI/CD):** Airflow, Step Functions, GitHub, Bitbucket.
- Storage:** Apache Iceberg, Parquet
- Core:** Performance Optimization, Data Quality, System Design

PROFESSIONAL EXPERIENCE

MDTHINK

Sr. Data Engineer (AWS Systems Engineer) – Python/PySpark/SQL

Mar 2024 - Present

Maryland

- Designed and scaled distributed ETL pipelines (Glue, PySpark, Redshift) supporting mission-critical analytics, reporting systems, and large-scale data processing across CJAMS, CSMS, and CME
- Owned end-to-end data platform (Glue, PySpark, Redshift, Iceberg) processing high-volume, large-scale datasets with production SLAs, ensuring reliability and scalability
- Enabled data-driven decision-making by delivering reliable reporting datasets, KPIs, and metrics for downstream analytics and business stakeholders
- Built and maintained curated warehouse tables (Gold layer) supporting finance, operations, and compliance reporting.
- Implemented automated data validation and reconciliation checks, ensuring high data accuracy and reliability for downstream users.
- Designed and implemented lakehouse architecture (Bronze–Silver–Gold) across 100+ datasets, enabling scalable analytics, reporting, and consistent data consumption
- Built high-performance ETL pipelines with CDC, reducing latency by 30–40% and improving data availability for analytics and reporting use cases
- Designed dimensional data models, KPIs, and reporting layers to support analytics and dashboards
- Optimized Spark and Redshift workloads, improving query performance and reducing compute cost for large-scale data processing and analytics
- Implemented secure and governed data access layer (S3, Lake Formation, IAM), enabling controlled and compliant data consumption
- Designed event-driven data pipelines using Lambda, SQS, SNS, Kinesis, and Step Functions for scalable and reliable data processing
- Built data quality and validation frameworks, improving data reliability, consistency, and trust for downstream reporting and analytics

- Led CI/CD automation for ETL pipelines, improving deployment speed, reliability, and operational efficiency
- Translated ambiguous business requirements into scalable data models and pipelines, enabling consistent analytics solutions
- Analyzed and resolved data pipeline performance and reliability issues, identifying root causes and implementing scalable fixes
- Developed observability framework (CloudWatch) for monitoring, alerting, and reducing incident resolution time
- Ensured data accuracy, integrity, and auditability for downstream reporting systems

Macquarie

Aug 2021 - Jan 2024

Data Engineer (AWS) - Python

PA

- Built data lake and data warehouse platform (S3, Glue, Redshift) for TB-scale financial datasets, enabling analytics and reporting
- Built and maintained analytical data models and reporting datasets, enabling business users to derive insights from large-scale financial data
- Developed scalable ETL pipelines (Spark, Python) for large-scale data ingestion, transformation, and processing
- Designed dimensional data models (star schema) to support financial reporting, analytics, and business insights
- Improved SQL query performance by 50%+ through partitioning, schema optimization, and SQL performance tuning
- Enabled self-service analytics and reporting using Athena and Redshift Spectrum
- Automated workflows using Step Functions, reducing manual intervention by 80% and improving operational efficiency
- Partnered with stakeholders to translate requirements into data models, metrics, and reporting datasets
- Built serverless data processing solutions using Lambda, DynamoDB, and S3 for scalable and reliable data workflows

Yokogawa Pvt Ltd

Aug 2017 - Jan 2020

Python Developer

Bangalore, India

- Developed Python-based data transformation and processing workflows to support data ingestion and analytics
- Built data validation and cleaning pipelines to improve data quality and consistency
- Performed data normalization and aggregation using Python libraries (pandas, NumPy)

EDUCATION

University of Colorado Denver

2021

Master of Science in Computer Engineering

GPA: 3.71

PROFESSIONAL CERTIFICATIONS

- **AWS Certified Solutions Architect – Professional (2026)**
- **AWS Certified Data Engineer – Associate: Issued by AWS (May 2025)**
- **AWS Certified AI Practitioner: Issued by AWS (May 2025)**
- **AWS Certified Solutions Architect Associate (SAA – C03): Issued by AWS (June 2023)**
- **Apache Airflow: Issued by Astronomer (August 2021)**
- **Databricks Lakehouse Platform Accreditation: Issued by Databricks (August 2023)**